One Outcome from 330: A Theorem on Folding Eleven Nested Dimensions into a 3D + 1

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Abstract

Of the C(11, 4) = 330 ways to choose four of eleven objects, counting alone prefers none. We ask what remains of this symmetry when the eleven objects are nested dimensions, indexed 1 to 11, and the four chosen are the leading dimensions of a 3D + 1: three equivalent axes at 90° forming a cube, together with a fourth axis.

Nested is used in two standard senses, as an order, which gives each dimension its index as a weight, and as a projection to a lower-dimensional space, which groups the eleven behind four leaders.

We define the folding of the nested eleven into a 3D + 1 by three conditions, on the geometry of the cube, on the weights, and on the leaders, and prove that exactly one of the 330 choices satisfies them.

The leading dimensions are 1, 2, 3, and 5. The folding is the cube of 27 positions, with three axes of weight 19 and a fourth axis of weight 9, and the dimension 4 never leads. The cubes of 1 and 8 are impossible at every depth, and the cube of 27, the smallest that fits, requires exactly eleven nested dimensions.

The proofs are elementary, and every case is confirmed by exhaustive computer search. The relation of the result to eleven-dimensional physics is discussed in the final section.

Keywords: nested dimensions, eleven dimensions, 3D + 1, folding, cube of 27, index weights, projection, C(11, 4)

The question

When a 3D + 1 is chosen from eleven nested dimensions, what outcomes are possible?

Choosing four of eleven objects can be done in C(11, 4) = 330 ways, and as plain counting all 330 choices are equal. This note asks what is left of the 330 when the eleven are dimensions, nested one in another, and the four are the three axes of a cube together with a fourth axis. The answer is that one outcome is possible, and the note proves it.

The standard definitions used

Every input of the note is a standard object.

- **A nested set of dimensions** is an ordered chain of spaces

$$V_1 \subset V_2 \subset \ldots \subset V_{11},$$

each inside the next. Linear algebra calls it a flag [1]. The index of V_k is its place k in the chain.

- **A 3D space** is Euclidean three-space: three mutually orthogonal axes at 90°, with the cube as its representation [2].
- **Seeing a space from a higher one** is a projection, a map from a larger space onto a smaller one [2].

Two ways of reading "nested"

The word nested has two meanings, and both are used here.

Nested as an order. Like a set of Russian dolls, each inside the next, the dimensions 1 to 11 sit one inside the other: 1 inside 2, 2 inside 3, and so on, so that n is higher than n − 1. This is the classical, mechanical view. It gives every dimension its place and its weight, its index, and it gives the whole the total weight 1 + 2 + … + 11 = 66.

The numbers are indices. The numbers 1 to 11 are used here as the indices of the eleven nested dimensions: the index of a dimension is its place in the nesting, and its value is the weight that place carries. They are not symbols of a counting base, and the written form does not matter. The eleventh dimension is the eleventh whether it is written 11 in base 10 or with a letter.

Nested as a view from higher dimensions. A shadow of a cube on a wall, or a photograph of a room, is a flat picture of something with more dimensions than the picture has. In the same way a 3D + 1 can be read as a space folded from eleven dimensions down to four. Behind each axis stands a group of the eleven, and one dimension of each group stands for it, its leading dimension. Most people do not picture a higher dimension behind what they see, but a mathematician works with both views, the order as a tool and the projection as geometry, and this note needs both.

Each reading supplies a different part of what the note uses.

Reading	What it supplies
Nested as an order	The index of each dimension as its weight, the sums of weights, and the total 66
Nested as a view from higher dimensions	The four groups, one behind each axis, and one leading dimension in each group

Where the two views meet

Each view alone leaves the choice open. The order alone allows any four of the eleven to be the leading dimensions, the 330 choices, and the projection alone, with the dimensions as unordered labels, has no weights and no leaders. Held together, the weights of the four leading dimensions must add to 11, the number of dimensions, and the weights of each group must fit the cube. Then one choice is left: the leaders 1, 2, 3 and 5, with axes of weight 19 and a left-over of weight 9.

The rest of the note works out what follows from holding the two views together, with the cube of Euclidean geometry as the shape of the 3D.

The example

Take the numbers 1 to 11. They are the indices of eleven nested dimensions, and the order matters: 2 is higher than 1, 3 is higher than 2, and so on. Their sum is 1 + 2 + … + 11 = 66.

Split them into three groups of three, called A, B and C, and leave two numbers over. The leading dimension of each group, the one that stands for it, is written first.

Group	Numbers	Sum
A	1, 7, 11	1 + 7 + 11 = 19
B	2, 8, 9	2 + 8 + 9 = 19
C	5, 4, 10	5 + 4 + 10 = 19
Left over	3, 6	3 + 6 = 9

The three groups have the same sum, 19, and the two left-over numbers have the sum 9.

Together 3 × 19 + 9 = 57 + 9 = 66, so every number from 1 to 11 is used once.

Add one group and the left-over: 19 + 9 = 28 = 27 + 1.

The groups are the three axes of a cube. Each axis has three dimensions, so the cube has 3 × 3 × 3 = 27 positions. The left-over two numbers add one more unit, the +1 of 3D + 1. The four leading dimensions are 1 in A, 2 in B, 5 in C and 3 in the left-over pair, and they add to 1 + 2 + 3 + 5 = 11, the number of dimensions.

A second split with the same sums shows that the example is one of several: the left-over {2, 7}, with groups {1, 8, 10}, {3, 5, 11} and {4, 6, 9}, each summing to 19, and 2 + 7 = 9. The section "The six splits with 19 and 9" below lists all six.

What a 3D is

In Euclidean geometry a 3D space is three axes at 90° to each other, and its representation is the cube. A cube can be read at three levels, by the number of points on each axis.

Points on each axis	Which points	Count
1	the centre only	1 × 1 × 1 = 1
2	the two ends of each axis, so the corners only	2 × 2 × 2 = 8
3	the two ends and the centre of each axis	3 × 3 × 3 = 27

In the cube of 27 every kind of point of a cube is present: 8 corners, 12 edge midpoints, 6 face centres and 1 centre, and 8 + 12 + 6 + 1 = 27. It is the smallest cube that has a centre together with the corners, the edges and the faces. The cube of 8 has corners and no centre, and the cube of 1 is the centre alone. So, 27 is the complete cube, in which every point is a point of the same Euclidean space.

In 11 nested dimensions, an axis of k dimensions has k points, so the cube of 27 uses three dimensions on each of the three axes, nine in all. The other two dimensions are the fourth axis, the +1 of 3D + 1.

The definition and the theorem

Definition (folding). The nested 11D carries the indices 1 to 11 as weights, with (n) higher than (n − 1). It is folded into a 3D + 1 when three conditions hold.

- **F1, the geometry.** Three equivalent axes at 90°, each of k dimensions, form a cube of k × k × k positions. The remaining dimensions are the fourth axis.
- **F2, the weights.** Equivalent axes have the same weight s, and the remaining dimensions have weight p. The weight of one axis plus the weight of the remainder accounts for the positions of the cube plus one, the one being the fourth axis: $s + p = k^3 + 1$.
- **F3, the leading dimensions.** Each of the three axes and the fourth axis has one leading dimension, the one that stands for its group, and the four leading dimensions add to 11, the number of dimensions.

F1 comes from the geometry of a 3D space, F2 and F3 from holding the order and the projection together. Since $3s + p = 1 + 2 + \ldots + 11 = 66$ always holds, F2 gives $2s = 65 - k^3$.

Which dimension of a group is its leader. The index alone does not say. If the lowest number of each group were taken as its leader, the four would add to 1 + 2 + 3 + 4 = 10, and 11 could not be reached. F3 asks for leading dimensions whose weights add to the depth, and the only four different positive numbers that add to 11 are 1, 2, 3, and 5. In this note the leader is written first in each group.

Consequence: 5 appears and 4 does not. Since the leaders are 1, 2, 3 and 5, the number 4 is never a leader. The dimension 4 is always carried inside the group of the leader 5, or as its partner in the pair. The space of the four leaders is four-dimensional, but it is not the first four dimensions.

Write V_k for the nested space of the first k dimensions. The space of the leaders contains V_3 and lies inside V_5, so it sits between the third and fifth nested spaces and is not V_4. The four-dimensional spaces between V_3 and V_5 form a line of choices with one parameter t, spanned by V_3 and $5 + t \cdot 4$. The space of the leaders is the point t = 0, and V_4 is the other end, $t = \infty$.

The Theorem.

Of the C(11, 4) = 330 ways to choose four of the eleven nested dimensions as the leading dimensions, exactly one allows the nested 11D to fold into a 3D + 1.

The folding it allows has one shape: the cube of 27 positions, with three dimensions on each of its three axes, each axis of weight 19, and a left-over of two dimensions of weight 9.

The four leading dimensions are 1, 2, 3, and 5, so dimension 4 never leads.

The Proof.

1. **From F1.** With k dimensions on each axis the three axes use 3k of the eleven dimensions. Since 3k is at most 11, k is 1, 2 or 3, and the cube has 1, 8 or 27 positions. A cube with k = 4 needs 12 dimensions and does not fit.
2. **From the nested order.** The three equivalent axes have the same weight s and the remainder has weight p, so 3s + p = 66.
3. **From F2.** $s + p = k^3 + 1$, and with step 2, $2s = 65 - k^3$.
4. **The three cubes.** For k = 1, each axis is one dimension, and three different indices cannot have the same weight. For k = 2, 2s = 65 − 8 = 57 is odd, so s is not a whole number. For k = 3, 2s = 65 − 27 = 38, so s = 19 and p = 66 − 57 = 9. Only the cube of 27 is possible, with axes of weight 19 and a leftover of weight 9.
5. **From F3.** The four leaders add to 11, and the only four different positive numbers that add to 11 are 1, 2, 3, and 5. This is one of the 330 choices of four, and the others are excluded. The groups exist: the example has the groups {1, 7, 11}, {2, 8, 9} and {5, 4, 10} and the left-over {3, 6}, and the six splits with weights 19 and 9 are listed below.

So, the folding leaves the cube of 27, the weights 19 and 9, and the leading dimensions 1, 2, 3, and 5. ■ *(every case was also checked by computer search over all splits of 1 to 11)*

Why only the cube of 27

A cube in a Euclidean space has three axes at 90°, and with k dimensions on each axis it has k × k × k positions. In 11 dimensions the possible cubes are 1, 8 and 27.

Cube	D per axis	Left over	Equal-sum splits of 1 to 11	With s + p = cube + 1
1 × 1 × 1 = 1	1	8	0	0
2 × 2 × 2 = 8	2	5	50	0
3 × 3 × 3 = 27	3	2	32	6

The cube of 1. Each axis holds one dimension, so the three axis sums are three different index numbers. Three different numbers cannot be equal, so no equal split exists.

The cube of 8. Equal splits exist, 50 of them, for example the pairs {1, 11}, {2, 10} and {3, 9}, each summing to 12. But F2 needs s + p = 8 + 1 = 9. With 3s + p = 66 this gives 2s = 57, and 57 is odd, so s is not a whole number. No split can meet the condition.

The cube of 27. 2s = 65 − 27 = 38 gives s = 19, a whole number, and the splits exist. The counts in the table are from an exhaustive computer search.

So, among the cubes that fit in 11 dimensions, only the cube of 27 has a 3D + 1.

Why 19 only

In the cube of 27, three equal triples can have other sums than 19. An exhaustive search of all splits of 1 to 11 into three equal-sum triples and a left-over pair finds 32 splits, spread over seven values of s. The condition s + p = 28 keeps only one of them.

Triple sum s	Left-over p	s + p	Number of splits	Meets s + p = 28?
15	21	36	2	no
16	18	34	4	no
17	15	32	6	no
18	12	30	8	no
19	**9**	**28**	**6**	**yes**
20	6	26	4	no
21	3	24	2	no

The value of s + p falls by 2 at every step, since s + p = 66 − 2s, so 28 is reached exactly once, at s = 19.

An example of a split that does not meet the condition: the left-over {1, 11} with the triples {2, 6, 10}, {3, 7, 8} and {4, 5, 9}, each summing to 18. The leading four 1, 2, 3 and 5 still add to 11, but 18 + 12 = 30, not 28. So F2 is what selects 19.

A fact about the order. In each of the six splits with s = 19, the lowest number of each of the three groups together with the lowest of the left-over pair is 1, 2, 3, and 4, which add to 10. To get 11, the 4 must be replaced by the 5, and that is possible only if 4 and 5 are in the same group.

The six splits with 19 and 9

There are six splits with the triple sum 19 and the left-over sum 9. They are the same cube numerically: each axis weighs 19, the left-over weighs 9, and each of the 27 positions is the same kind of position. They differ only in which index numbers sit on which axis.

Left-over pair	Group 1	Group 2	Group 3	Leading four adding to 11
{1, 8}	{2, 6, 11}	{3, 7, 9}	{5, 4, 10}	1, 2, 3, 5
{1, 8}	{2, 7, 10}	{3, 5, 11}	{4, 6, 9}	none
{2, 7}	{1, 8, 10}	{3, 5, 11}	{4, 6, 9}	none
{3, 6}	{1, 7, 11}	{2, 8, 9}	{5, 4, 10}	1, 2, 3, 5
{5, 4}	{1, 7, 11}	{2, 8, 9}	{3, 6, 10}	1, 2, 3, 5
{5, 4}	{1, 8, 10}	{2, 6, 11}	{3, 7, 9}	1, 2, 3, 5

The only four different positive numbers that add to 11 are 1, 2, 3 and 5, so a leading four adding to 11 is always this set when it exists. It exists in four of the six splits, those where 4 and 5 are in the same group.

The left-over pair and the centre of the nesting. Among the four splits that keep the leading four at 11, the left-over pair is {1, 8}, {3, 6} or {5, 4}.

The map i → 12 − i, which reads the nesting from the other end, fixes one index only, 6, the centre of 1 to 11. Only the pair {3, 6} contains it.

The pair {1, 8} lies at the extreme end of the nesting, and the pair {5, 4} lies wholly in its lower half.

The cube of 27 has a centre point, and in this split the fourth axis carries the centre of the nesting.

This is an observation about the numbers. It is a further condition, and it chooses one of the four.

What the numbers do not choose.

- **Which two numbers are left over.** F1 to F3 fix the weights 19 and 9. Each of the four splits is the same cube, and the numbers do not prefer one of them without the further condition above.
- **Reversal.** The map i → 12 − i takes the split with 19 and 9 to a split with 17 and 15, for example the left-over {6, 9}. That split has s + p = 32, so it does not meet F2, and the reversal does not give a second 3D + 1.

The weight of the fourth axis. If F2 is replaced by the geometric statement that the fourth axis weighs as much as one face of the cube, $p = k^2$, then $3s + p = 66$ gives for k = 3 the weights p = 9 and s = 19, and at depth 11 the same six splits follow. For k = 1 and k = 2 no solution exists at depth 11. This statement does not fix the depth: at depth 12 it also gives splits, five of them. The depth 11 follows from F2.

From the choices to one

Each condition removes choices, and the counts show where.

Choices	Before	Condition	After
Four chosen dimensions of eleven	330	the four add to 11 (F3)	1
Cubes that fit in 11 dimensions	3	$s + p = k^3 + 1$ (F2)	1
Equal splits of 1 to 11 with cube 27	32	$s + p = 28$ (F2)	6
The six splits	6	the leading four add to 11 (F3)	4
The four splits	4	the left-over contains the centre index 6	1

The first three rows are the theorem. The last two show what remains among the arrangements of the same weights.

Why 11

Build up from the smallest cube and ask at which depth N of nested dimensions it first fits. The index numbers 1 to N add to $T = N(N + 1) / 2$. The two conditions are $3s + p = T$ and $s + p = k^3 + 1$, so $s = (T - k^3 - 1) / 2$ and $p = k^3 + 1 - s$.

- **k = 1, the cube of 1.** Each axis is one dimension, and three different index numbers cannot have the same sum. It is impossible at every depth.
- **k = 2, the cube of 8.** The three axes are three pairs of different numbers, six numbers in all, so 3s is at least 1 + 2 + … + 6 = 21 and s is at least 7. Since p = 9 − s cannot be negative, s is 7, 8 or 9, and T = 3s + p = 2s + 9 is 23, 25 or 27. None of these is a sum 1 + … + N. It is impossible at every depth.
- **k = 3, the cube of 27.** The axes use nine numbers, so N is at least 9. Here p = 28 − s and T = 2s + 28, which gives p = 42 − N(N + 1) / 4. For N = 9 and N = 10 this is 19.5 and 14.5, not whole numbers. For N = 11 it is 9. For N = 12 it is 3, but the left-over then has three numbers whose sum is at least 6. For N = 13 and above it is negative. So, N = 11 is the only depth.

The smallest cube that has a 3D + 1 is therefore the cube of 27, with three dimensions on each axis, and it needs exactly 11 nested dimensions: the smallest cube that fits sets the depth. The cube of 27 is the first odd cube above 1, so it is the first cube with a centre point, and it is the only one of the three that fits.

A cube with four dimensions per axis, 64 positions, needs 12 dimensions for its three axes and one more for the fourth axis, so N = 13. Then T = 91, 2s = 91 − 65 = 26, s = 13 and p = 52, and one left-over index cannot be 52. With two left-over indices, N = 14, T = 105, s = 20 and p = 45, and two indices cannot reach 45. So, the cube of 64 does not fit with a small fourth axis. At depth 17 the cube of 64 does have solutions, 194 splits, but with a fourth axis of five dimensions and not one or two. The theorem of this note concerns eleven nested dimensions, and it is not a statement about other depths.

From 330 to 1. There are C(11, 4) = 330 ways to choose four of the eleven indices. Because the indices are nested, each one is a weight and the order matters, and F3 asks that the four leaders add to 11. The only four different positive numbers that add to 11 are 1, 2, 3 and 5, so exactly one of the 330 choices is allowed. At depth 10 the same condition would give 1, 2, 3 and 4, but the cube of 27 does not fit at depth 10, as shown above.

Relation to physics

Eleven dimensions are already used in mainstream physics: eleven-dimensional supergravity [3] and M-theory [4], in which the dimensions beyond those that give a world of three space dimensions and one of time are folded away.

The usual split is four dimensions and seven hidden ones.

The split in this note is different: three axes of three dimensions and a fourth axis of two, 3 × 3 + 2 = 11, with the leading dimensions 1, 2, 3 and 5.

The 3D + 1 of the theorem is the space in which matter takes its shape, and the nested eleven is what it is folded from, so the theorem describes one thing and not two. Whether the space we live in is folded from such a nested 11D is the physical hypothesis. For that case the theorem says what the folding must be: leaders 1, 2, 3, and 5, axes of weight 19, and a left-over of weight 9.

In the standard compactifications the four extended dimensions are a chosen factor of the eleven, and the theory does not say which of the eleven they are or what a 3D is made of. For the nested eleven the

theorem answers both: the 3D is nine dimensions, three on each of three axes, and the fourth axis is two dimensions.

Closing

Of 330 choices that counting alone cannot tell apart, the nested order and the geometry of a cube leave one. Eleven nested dimensions fold into a 3D + 1 in exactly one way: leaders 1, 2, 3 and 5, three axes of three dimensions each of weight 19, a fourth axis of two dimensions of weight 9, a cube of 27 positions. No smaller cube can be folded at any depth, and the cube of 27 cannot be folded at any depth but eleven. The dimension 4 never leads.

This is a theorem about the nested eleven and the cube, proved from three stated conditions, and every step can be checked by hand. If the space in which matter takes its shape is folded from nested dimensions, this is the form that folding must take, and it differs from the usual split into four and seven, so the two can be compared.

The numbers fix the weights 19 and 9. They do not fix which pair is left over, and among the four splits that keep the leaders at 11, the one that carries the centre index 6 is noted as an observation. Larger cubes at greater depths are not treated here.

Computation

Every count in this note is reproduced by one script, nested_cube_checks.py, which uses only the Python standard library and runs in under a second. The script and its output are published with this note on Zenodo and in the author's GitHub repository: the script at https://github.com/quranquantum/quranquantum/blob/main/nested_cube_folding/nested_cube_checks.py and its output at https://github.com/quranquantum/quranquantum/blob/main/nested_cube_folding/nested_cube_checks_output.txt. The table lists what it returns.

Statement	Result
Ways to choose four of eleven	C(11, 4) = 330
Four different indices from 1 to 11 adding to 11	(1, 2, 3, 5) only
Equal splits of 1 to 11 into three triples and a left-over pair	32, by s = 15 to 21: 2, 4, 6, 8, 6, 4, 2
Splits meeting F2 (s = 19, p = 9)	6
Of these six, splits with leaders adding to 11	4
Of these four, left-over containing the centre index 6	1, the pair {3, 6}
Cubes of 1, 8 and 27: equal splits	0, 50 and 32
Cubes of 1, 8 and 27: splits meeting F2	0, 0 and 6

Statement	Result
Depths at which the cube of 27 meets F2 (N up to 18)	11 only
Depths at which the cube of 64 meets F2 (N up to 18)	17 only, 194 splits, with a fourth axis of five dimensions
Fourth axis weighing one face, $p = k^2 = 9$: splits at depth 11 and 12	6 and 5

Output of nested_cube_checks.py. The full output of the script, as run, is given here.

```
======================================================================
1. Choosing four of eleven
======================================================================
C(11,4) = 330
four different indices from 1..11 adding to 11: [(1, 2, 3, 5)]

======================================================================
2. Equal splits of 1..11 into three triples and a left-over pair
======================================================================
total splits: 32
  s   p   s+p   splits
  15  21   36    2
  16  18   34    4
  17  15   32    6
  18  12   30    8
  19   9   28    6
  20   6   26    4
  21   3   24    2

======================================================================
3. F2 (s + p = 28) keeps s = 19; the six splits; F3 on the leaders
======================================================================
splits with s = 19, p = 9: 6
  left-over {8, 1} groups [{2, 11, 6}, {9, 3, 7}, {10, 4, 5}] -> leaders [(1, 2, 3,
5)]
  left-over {8, 1} groups [{2, 10, 7}, {11, 3, 5}, {9, 4, 6}] -> no leaders adding to
11
  left-over {2, 7} groups [{8, 1, 10}, {11, 3, 5}, {9, 4, 6}] -> no leaders adding to
11
  left-over {3, 6} groups [{1, 11, 7}, {8, 9, 2}, {10, 4, 5}] -> leaders [(1, 2, 3,
5)]
  left-over {4, 5} groups [{1, 11, 7}, {8, 9, 2}, {10, 3, 6}] -> leaders [(1, 2, 3,
5)]
  left-over {4, 5} groups [{8, 1, 10}, {2, 11, 6}, {9, 3, 7}] -> leaders [(1, 2, 3,
5)]
splits keeping a leader set that adds to 11: 4
of these, left-over contains the centre index 6: 1

======================================================================
4. The three cubes in 11 nested dimensions
======================================================================
```

```
k  cube  equal splits  splits meeting F2 (s+p=k^3+1)
1    1        0        0
2    8       50        0
3   27       32        6

======================================================================
5. Depth: for which N does a cube of k^3 meet F2 (remainder may have
   any number of indices)?  counts of splits
======================================================================
k=1 (cube 1): depths with a solution: none up to N=18
k=2 (cube 8): depths with a solution: none up to N=18
k=3 (cube 27): depths with a solution: [(11, 6)]
k=4 (cube 64): depths with a solution: [(17, 194)]

6. Cube of 27 formula check: p = 42 - N(N+1)/4
  N=9: p = 19.5
  N=10: p = 14.5
  N=11: p = 9.0
  N=12: p = 3.0
  N=13: p = -3.5
  N=14: p = -10.5

======================================================================
7. Alternative F2': the fourth axis weighs one face, p = k^2 = 9
======================================================================
  depth 9: 0 splits
  depth 10: 0 splits
  depth 11: 6 splits
  depth 12: 5 splits
  depth 13: 0 splits
  depth 14: 0 splits
  depth 15: 0 splits
  depth 16: 0 splits
```


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